Week Summary

Introduction to MPI

MPI summary

- **MPI basics:** setting up and shutting down the MPI environment, determining the number of processes and process rank, building and running an MPI program
- **Point-to-point communication:** communication between a pair of MPI processes
- **Collective communication:** communication within a group of MPI processes
- **Blocking and non-blocking communication**
- **Domain decomposition:** distributing work in an MPI program

OpenMP vs. MPI

OpenMP	MPI
Single instance of program which forks threads in parallel region	Multiple copies of program which communicate by passing messages
Add compiler directives	Make calls to MPI library functions
Part of the compiler (with a run time library)	External library (but normally compiled with a wrapper script)
Shared memory only	Distributed memory and shared memory
Suitable for incremental parallelisation	Incremental parallelism is harder (better to design from the start)

MPI: advantages and disadvantages

- A significant advantage over OpenMP is that we can use MPI when processor cores do not share memory
- MPI requires more work than OpenMP - we have to distribute the work ourselves (e.g. domain decomposition)
- MPI is less suited to incremental parallelism - best if designed in from the start